Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

AutoCAD Masterclass

Course Code: C391

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 1.0

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DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
1.0	1 October 2026	First release of the AutoCAD Masterclass: five topics and 16 hands-on labs over two days, 9:30am – 5:30pm (7.5 instructional hours per day).	Dr Alfred Ang

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Course Information

Course Title	AutoCAD Masterclass
Course Code	C391
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	2 days · 7.5 instructional hours per day (15 hours)
Daily Timing	9:30 am – 5:30 pm (30-min lunch)
Mode	Instructor-led, hands-on AutoCAD drafting labs — one computer with AutoCAD per learner
Trainer	Dr Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Read and visualize AutoCAD technical drawings
- LO2: Produce 2D and 3D technical drawings using AutoCAD
- LO3: Create part lists using AutoCAD
- LO4: Create AutoCAD design documentation
- LO5: Produce design specification using AutoCAD

Course Schedule

Day 1 — AutoCAD Foundations, Drawing & Modifying Objects

Time	Duration	Topic / Activity	Slides
9:30–10:00	30 min	Welcome, course introduction, ground rules, trainer and learner introductions, learning outcomes and course outline	Slides 3–12
10:00–10:45	45 min	**Topic 1 — Getting Started with AutoCAD: user interface, workspaces, model/paper space, navigation, precision aids, units (concepts + demo)**	Slides 13–24
10:45–11:15	30 min	Hands-on: Lab 1: Setting Up a New Drawing	Slides 25–27

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Time	Duration	Topic / Activity	Slides
11:15–12:15	60 min	**Topic 2 — Technical Drawing with AutoCAD: lines, circles, rectangles, polygons, ellipses, arcs, polylines, splines, construction geometry, isometric drawing (concepts + demo)**	Slides 28–36
12:15–12:45	30 min	Lunch break	
12:45–14:45	120 min	Hands-on: Lab 2: Creating Lines and Coordinates; Lab 3: Circles, Rectangles and Polygons; Lab 4: Polylines, Arcs, Ellipses and Splines; Lab 5: 2D Isometric Drawing	Slides 37–45
14:45–15:30	45 min	**Topic 3 — Modifying Objects: move/copy/rotate/offset, mirror/stretch/scale, trim/extend, fillet/chamfer, arrays (concepts + demo)**	Slides 46–55
15:30–17:15	105 min	Hands-on: Lab 6: Move, Copy, Rotate and Offset; Lab 7: Trim, Extend, Fillet and Chamfer; Lab 8: Mirror, Stretch, Scale and Arrays	Slides 56–62
17:15–17:30	15 min	Day 1 recap and Q&A	

Day total: 7.5 instructional hours (9:30am – 5:30pm, 30-min lunch).

Day 2 — Organizing Objects & Design Documentation

Time	Duration	Topic / Activity	Slides
9:30–9:45	15 min	Day 1 recap	
9:45–10:50	65 min	**Topic 4 — Organizing Objects: object properties, layers, grips, hide/isolate, blocks & attributes, external references (concepts + demo)**	Slides 63–71
10:50–12:05	75 min	Hands-on: Lab 9: Object Properties and Layers; Lab 10: Grips, Hide and Isolate	Slides 72–75

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Time	Duration	Topic / Activity	Slides
12:05–13:25	80 min	Hands-on: Lab 11: Blocks, Attributes and Part Lists; Lab 12: External References and PDF Import	Slides 76–80
13:25–13:55	30 min	Lunch break	
13:55–15:00	65 min	**Topic 5 — Design Documentation: text & styles, dimensions, multileaders, tables & part lists, layouts, viewports, plotting, hatching (concepts + demo)**	Slides 81–91
15:00–16:15	75 min	Hands-on: Lab 13: Text and Text Styles; Lab 14: Dimensions, Multileaders and Part-List Tables	Slides 92–95
16:15–17:15	60 min	Hands-on: Lab 15: Hatching and Detail Finishing; Lab 16: Layouts, Viewports and Plotting	Slides 96–100
17:15–17:30	15 min	Course recap, Q&A and course feedback	Slides 101–105

Day total: 7.5 instructional hours (9:30am – 5:30pm, 30-min lunch).

Lab Reference

Topic	Labs
Topic 01: Getting Started with AutoCAD	Lab 1
Topic 02: Technical Drawing with AutoCAD	Lab 2, Lab 3, Lab 4, Lab 5
Topic 03: Modifying Objects	Lab 6, Lab 7, Lab 8
Topic 04: Organizing Objects	Lab 9, Lab 10, Lab 11, Lab 12
Topic 05: Design Documentation	Lab 13, Lab 14, Lab 15, Lab 16

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